Google Advanced Data Analytics Capstone

Salifort Motors Employee Retention Predictive Analytics Project

Prepared for senior leadership and Human Resources stakeholders

Written by Brian McCarthy

Deliverable	Status
PACE strategy document	Completed with Plan, Analyze, Construct, and Execute responses
Capstone lab evidence	Python EDA, cleaning, model building, model evaluation, and interpretation
Executive summary	Completed one-page stakeholder summary with insights, limitations, and next steps
Dataset used	archive.zip / HR_comma_sep.csv; 14,999 original rows, 10 columns

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1. Project Overview and Business Problem

Salifort Motors is experiencing employee turnover that creates operational risk, higher recruiting costs, training costs, lost productivity, and potential morale issues. The Human Resources team collected employee survey and employment data and asked for data-driven suggestions to identify factors that make employees likely to leave.

The project objective is to use Python for exploratory data analysis, data cleaning, visualization, predictive modeling, and stakeholder communication. The business question is: Which factors are most associated with employee departure, and how can Salifort Motors improve retention?

The chosen modeling approach includes a statistical baseline using logistic regression and machine learning models using decision tree and random forest classifiers. The random forest model is selected as the preferred operational model because it offers the best balance of precision, recall, F1-score, ROC AUC, and business interpretability through feature importance.

Business value:

- Identify employees who may be at high risk of leaving before turnover occurs.
- Prioritize retention initiatives around workload, tenure, satisfaction, promotions, and workplace safety.
- Help HR and leadership reduce replacement costs and improve employee experience.
- Translate model findings into practical business actions rather than only reporting technical scores.

2. Source Files and Deliverables Used

File / Input	How it was used
Pasted text(168).txt	Capstone scenario, project requirements, expected deliverables, HR dataset descriptions, and executive summary requirements.
Activity Templates_ Executive summaries.pdf	Executive summary layout guidance; used as inspiration for the one-page stakeholder summary sections: overview, objective, results, and next steps.
Activity Template_ Course 6 PACE strategy document.pdf	PACE document structure; used to organize Plan, Analyze, Construct, and Execute answers.
Pasted text (2)(7).txt	Capstone lab instructions, lab workflow, model assumptions, coding tasks, and reflection questions.
archive.zip / HR_comma_sep.csv	Employee dataset used for EDA, cleaning, model training, and model evaluation.

The dataset contains employee-reported satisfaction level, last evaluation, number of projects, average monthly hours, tenure, work accident flag, turnover flag, recent promotion flag, department, and salary band. The original CSV used the column names average_montly_hours, Work_accident, and Department, which were standardized to average_monthly_hours, work_accident, and department during cleaning.

3. Completed PACE Strategy Document

PACE: Plan Stage

Question	Completed response
Audience / stakeholders	Senior leadership, Human Resources, people managers, department heads, and workforce planning stakeholders at Salifort Motors.
Problem to solve	Predict whether an employee is likely to leave and identify the strongest factors associated with turnover so HR can improve retention.
Business impact	Improved retention can reduce hiring, onboarding, and training costs while improving employee satisfaction, continuity, and productivity.
Required resources	HR_capstone_dataset / HR_comma_sep.csv, Python, pandas, NumPy, scikit-learn, matplotlib, project instructions, PACE template, and executive summary template.
Deliverables	Cleaned EDA outputs, model comparison, selected model, performance metrics, feature importance, recommendations, executive summary, and code appendix.
Ethical considerations	Avoid punitive use of predictions. Use model output as a decision-support signal, not as an automated employment decision. Monitor bias by department, salary, tenure, and other categories.

PACE: Analyze Stage

Question	Completed response
EDA steps	Checked shape, data types, missing values, duplicates, descriptive statistics, outliers, target balance, group differences, and variable relationships.
Relevant variables	satisfaction_level, number_project, average_monthly_hours, time_spend_company, last_evaluation, promotion_last_5years, work_accident, salary, department, and left.
Data sufficiency	The dataset is sufficient for a first predictive model because it contains 11,991 unique employee records after duplicate removal and a clear binary target variable.
Transformations	Renamed columns to snake_case, removed exact duplicate rows, retained tenure outliers for tree-based models, one-hot encoded categorical variables, and scaled numeric variables for logistic regression.
Initial insights	Employees who left had lower average satisfaction, higher average monthly hours, longer tenure, fewer promotions, fewer work accidents recorded, and higher turnover among low salary bands.

PACE: Construct Stage

Question	Completed response
Prediction type	Binary classification: left = 1 if employee left and left = 0 if employee stayed.
Models built	Logistic regression baseline, decision tree classifier, and random forest classifier.
Independent variables	All available predictors except left: satisfaction, evaluation, projects, average monthly hours, tenure, work accident, recent promotion, department, and salary.
Model assumptions	Logistic regression target is categorical and sample size is sufficient. Tree-based models are appropriate because they handle non-linear interactions and are less sensitive to monotonicity and outliers.
Selection criteria	Accuracy, precision, recall, F1-score, ROC AUC, confusion matrix behavior, business usefulness, interpretability, and risk of false negatives.
Selected model	Random forest because it produced the strongest overall performance and clear feature importance for stakeholder action.

PACE: Execute Stage

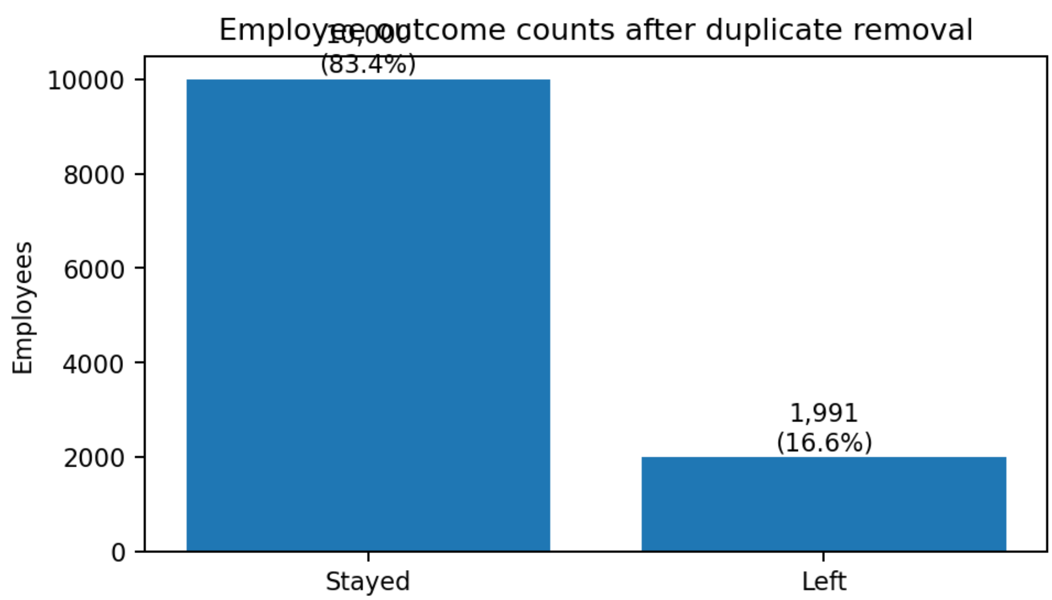
Question	Completed response
Key model insights	Satisfaction level, tenure, number of projects, average monthly hours, and last evaluation were the most predictive features. Salary and department were weaker but still useful context.
Recommendations	Reduce burnout risk, review workload thresholds, increase career development and promotion visibility, run stay interviews, and create retention plans for high-risk tenure groups.
Model improvement	Add richer HR data such as manager changes, commute, remote status, compensation history, performance trends, training participation, engagement survey comments, and exit interview reasons.
Ethics	Do not label employees as flight risks in a way that harms opportunities. Use aggregate and supportive interventions, audit subgroup performance, and involve HR governance before operational use.

4. Data Understanding and Cleaning

Metric	Value
Original rows	14,999
Original columns	10
Missing values	0
Exact duplicate rows removed	3,008
Rows after duplicate removal	11,991
Employees who stayed	10,000 (83.4%)
Employees who left	1,991 (16.6%)
Tenure outlier rows	824

Cleaning decisions:

- Standardized column names to snake_case for consistency and cleaner Python code.
- Removed exact duplicates to avoid repeated employee records influencing model training.
- Did not remove tenure outliers for the selected random forest because tree-based models can handle such values better than linear models and tenure extremes may carry retention information.
- Kept all available predictive variables because the business question requires identifying both workload and organizational factors associated with departure.



5. Exploratory Data Analysis

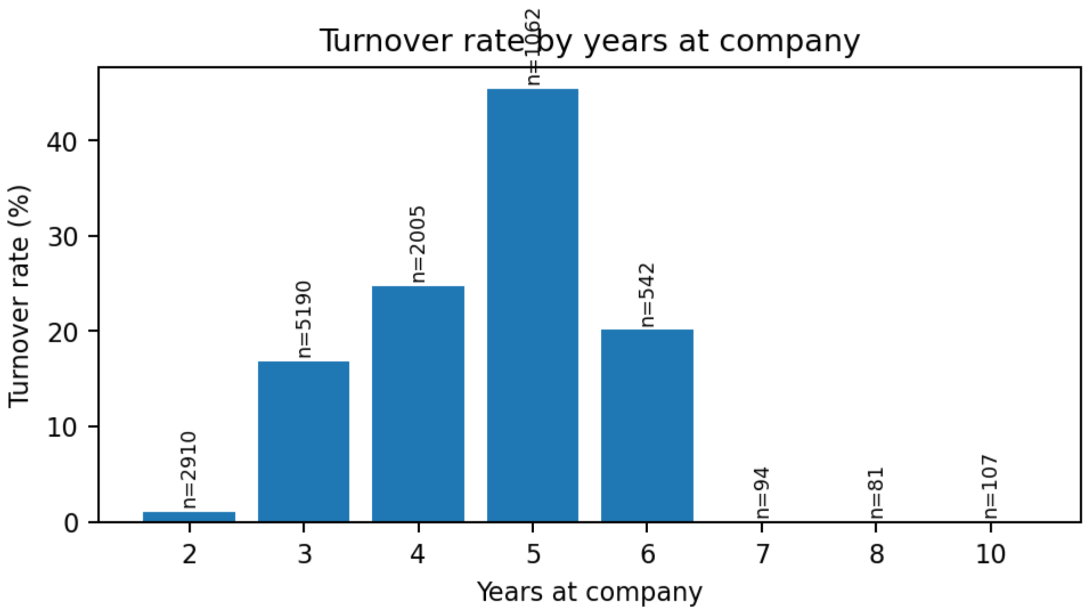
Average predictor values by employee outcome

Variable	Stayed average	Left average
satisfaction_level	0.667	0.440
last_evaluation	0.716	0.722
number_project	3.787	3.883
average_monthly_hours	198.943	208.162
time_spend_company	3.262	3.881
promotion_last_5years	0.019	0.004
work_accident	0.174	0.053

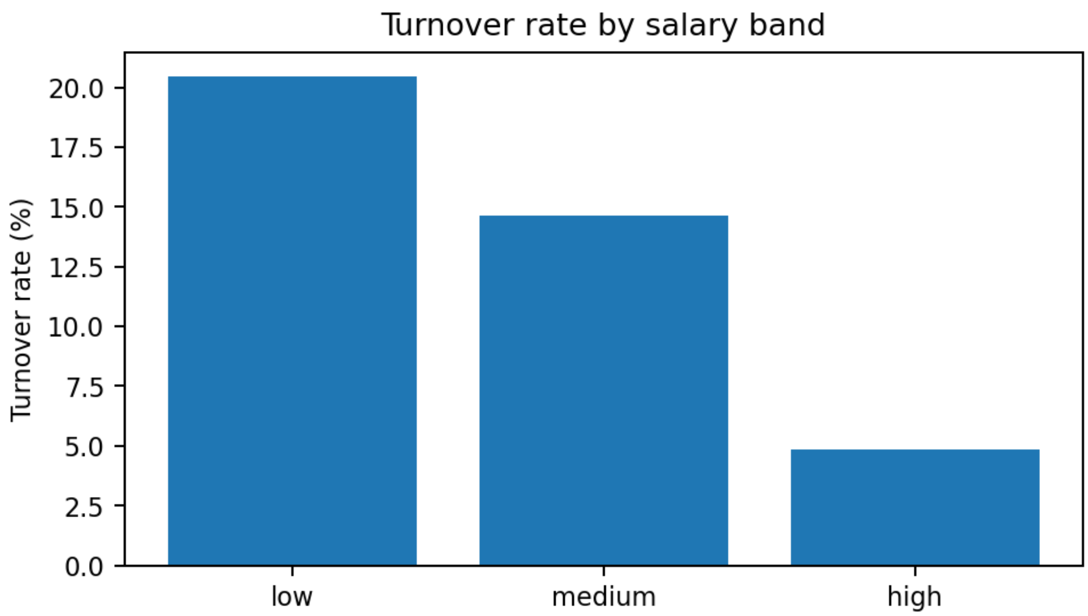
EDA interpretation:

- Employees who left had much lower mean satisfaction (0.440) than employees who stayed (0.667).
- Employees who left worked more monthly hours on average (208.2) than employees who stayed (198.9), which suggests workload pressure is relevant.
- Employees who left had longer average tenure (3.88 years) than employees who stayed (3.26 years), which may indicate a mid-tenure retention risk.
- Promotion in the last five years was uncommon overall, but especially rare among employees who left.
- Low salary employees had the highest turnover rate at about 20.5%; high salary employees had the lowest turnover rate at about 4.8%.

EDA Visualizations



EDA Visualizations Continued

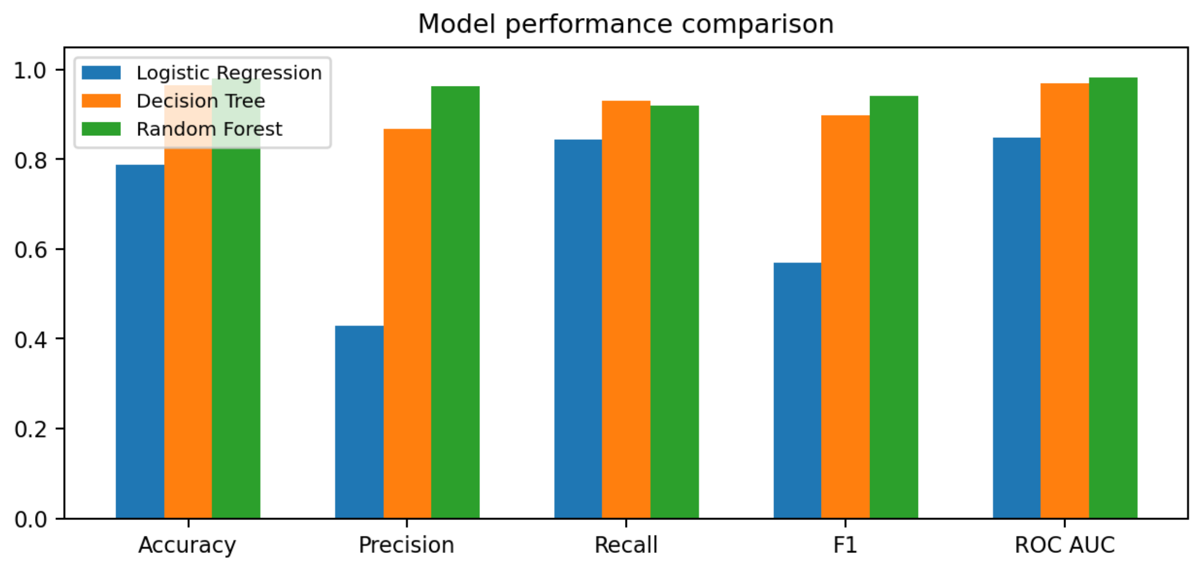


The visual evidence indicates that satisfaction, tenure, workload, and salary should be investigated closely. For HR stakeholders, the most actionable variables are workload balance, employee satisfaction, career progression, and compensation strategy.

6. Model Building and Evaluation

The dataset was split into training and testing sets using stratified sampling to preserve the stayed/left ratio. Numeric predictors were scaled for logistic regression, categorical variables were one-hot encoded, and all models were evaluated on the same test split.

Model	Accuracy	Precision	Recall	F1-score	ROC AUC	Confusion matrix [TN, FP; FN, TP]
Logistic Regression	0.788	0.429	0.844	0.569	0.849	[[1554, 447], [62, 336]]
Decision Tree	0.965	0.869	0.930	0.898	0.969	[[1945, 56], [28, 370]]
Random Forest	0.981	0.963	0.920	0.941	0.982	[[1987, 14], [32, 366]]



Selected Model: Random Forest

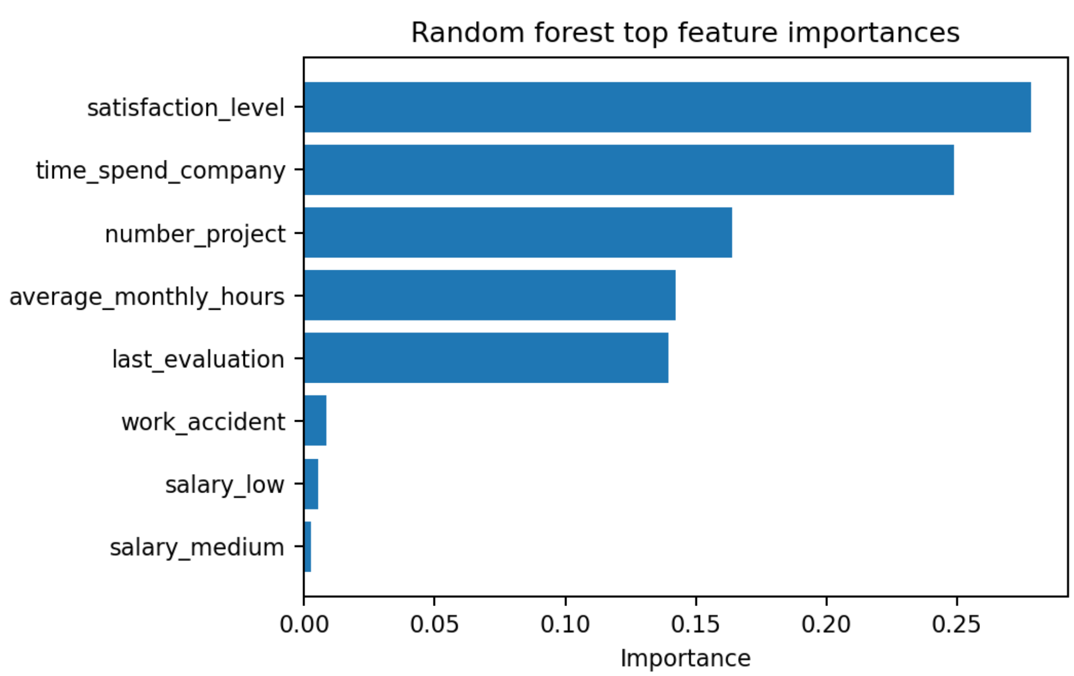
The random forest model is selected because it achieved the strongest overall results: accuracy 0.981, precision 0.963, recall 0.920, F1-score 0.941, and ROC AUC 0.982. In business terms, the model correctly identifies most employees likely to leave while keeping false alarms relatively low.

Confusion matrix interpretation for random forest:

Prediction result	Count	Meaning
True negatives	1,987	Stayed employees correctly predicted as stayed.
False positives	14	Stayed employees incorrectly flagged as likely to leave.
False negatives	32	Employees who left but were predicted to stay.
True positives	366	Employees who left correctly predicted as likely to leave.

Top predictive features:

Rank	Feature	Importance
1	satisfaction_level	0.279
2	time_spend_company	0.249
3	number_project	0.164
4	average_monthly_hours	0.142
5	last_evaluation	0.139
6	work_accident	0.008
7	salary_low	0.005
8	salary_medium	0.003
9	department_sales	0.002
10	department_technical	0.002



7. Executive Summary for Stakeholders

Section	Stakeholder-ready content
Overview	Salifort Motors can use employee data to identify turnover risk and focus retention actions on satisfaction, tenure, workload, career growth, and compensation.
Objective	Build and evaluate a predictive model to determine whether an employee is likely to leave, then convert model findings into practical HR actions.
Results	The random forest model performed best with 98.1% accuracy, 96.3% precision, 92.0% recall, 94.1% F1-score, and 98.2% ROC AUC on the held-out test set.
Key insights	The strongest predictors were satisfaction level, time at company, number of projects, average monthly hours, and last evaluation. Low satisfaction and high workload patterns appear especially important.
Benefits	The model can help HR prioritize retention conversations, workload balancing, manager coaching, and targeted career development.
Limitations	The model uses historical survey/employment data, does not prove causation, may miss unobserved factors, and should not be used for automated employment decisions.
Next steps	Create an HR retention dashboard, audit model fairness, conduct stay interviews, define workload thresholds, and pilot interventions with departments experiencing higher turnover.

8. Recommendations, Next Steps, and Ethical Considerations

Recommended business actions

- Workload management: review employees with very high average monthly hours or high project counts to identify burnout risk.
- Employee satisfaction interventions: conduct targeted stay interviews and manager check-ins for teams showing low satisfaction levels.
- Career growth: review promotion practices and create transparent development paths, especially for employees with several years of tenure.
- Compensation review: assess low salary bands and departments with above-average turnover for market competitiveness and internal equity.
- Operational monitoring: refresh the model quarterly and pair predictions with HR judgment, not automatic decisions.

Ethical safeguards

- Use predictions to offer support, coaching, resources, and retention outreach; do not use them to punish or limit opportunities.
- Audit performance by department, salary band, and tenure group to check whether false positives or false negatives concentrate in particular subgroups.
- Explain that the model estimates risk from historical patterns and does not determine why a specific individual will leave.
- Restrict access to individual-level predictions and share aggregate trends with broader leadership unless there is a legitimate HR purpose.
- Document data lineage, model versioning, feature definitions, validation metrics, and stakeholder approvals.

9. Python Code Appendix

```
# Google Advanced Data Analytics Capstone - Salifort Motors
# Written by Brian McCarthy

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score, confusion_matrix

# Load and clean data
df0 = pd.read_csv('HR_comma_sep.csv')
df = df0.rename(columns={
    'average_montly_hours': 'average_monthly_hours',
    'Work_accident': 'work_accident',
    'Department': 'department'
})
df_clean = df.drop_duplicates().copy()

# Target and predictors
X = df_clean.drop(columns=['left'])
y = df_clean['left']
cat_cols = ['department', 'salary']
num_cols = [c for c in X.columns if c not in cat_cols]

preprocess = ColumnTransformer([
    ('num', StandardScaler(), num_cols),
    ('cat', OneHotEncoder(drop='first', handle_unknown='ignore'), cat_cols)
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

models = {
    'Logistic Regression': Pipeline([
        ('preprocess', preprocess),
        ('model', LogisticRegression(max_iter=1000, class_weight='balanced'))
    ]),
    'Decision Tree': Pipeline([
        ('preprocess', preprocess),
        ('model', DecisionTreeClassifier(max_depth=8, min_samples_leaf=20,
                                         class_weight='balanced', random_state=42))
    ]),
    'Random Forest': Pipeline([
        ('preprocess', preprocess),
        ('model', RandomForestClassifier(n_estimators=120, max_depth=12,
                                         min_samples_leaf=5, class_weight='balanced',
                                         random_state=42, n_jobs=2))
    ])
}

for name, pipe in models.items():
    pipe.fit(X_train, y_train)
    y_pred = pipe.predict(X_test)
    y_prob = pipe.predict_proba(X_test)[:, 1]
    print(name)
    print('Accuracy:', accuracy_score(y_test, y_pred))
    print('Precision:', precision_score(y_test, y_pred))
    print('Recall:', recall_score(y_test, y_pred))
    print('F1:', f1_score(y_test, y_pred))
    print('ROC AUC:', roc_auc_score(y_test, y_prob))
    print('Confusion matrix:', confusion_matrix(y_test, y_pred))

# Feature importance for selected random forest model
rf = models['Random Forest']
ohe = rf.named_steps['preprocess'].named_transformers_['cat']
feature_names = num_cols + list(ohe.get_feature_names_out(cat_cols))
importances = rf.named_steps['model'].feature_importances_
feature_importance = pd.DataFrame({
    'feature': feature_names,
    'importance': importances
}).sort_values('importance', ascending=False)
print(feature_importance.head(10))
```

Appendix: Lab Reflection Answers

Plan reflection:

The primary stakeholders are Salifort Motors leadership, HR, and department managers. The objective is to understand what factors are associated with employee turnover and to build a model that predicts whether an employee will leave. Initial observations show a complete dataset with no missing values, a binary target, multiple useful workload and satisfaction predictors, and exact duplicates that should be removed. Ethical considerations include privacy, explainability, bias monitoring, and preventing punitive use of model output.

Analyze reflection:

EDA showed that employee departure is associated with lower satisfaction, higher workload, tenure effects, limited promotion history, and salary band differences. EDA is necessary before modeling because it validates data quality, reveals class balance, identifies outliers and duplicates, checks relationships, and guides feature engineering decisions.

Construct reflection:

The appropriate prediction task is binary classification. Logistic regression provides a baseline and interpretability, while decision tree and random forest models capture non-linear relationships. The random forest model performed best and is appropriate because the strongest signals appear to interact across satisfaction, tenure, workload, and evaluation variables.

Execute reflection:

The final recommendation is to use the model as a decision-support tool for HR retention planning. The model should be paired with human review, governance, and supportive employee programs. Future improvements should add richer HR data, retrain regularly, and evaluate subgroup fairness before operational deployment.